

What shape is a box made of primes?

Isoperimetric Quotients of Anisotropic Cuboids

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Abstract

We investigate the intrinsic shape of n -dimensional rectangular solids whose sides are drawn from consecutive integers (*oblong cuboids*) and consecutive primes (*puboids*). The isoperimetric quotient $M = S/V^{(n-1)/n} = 2n \cdot G/H$, where G and H are the geometric and harmonic means of the side lengths, provides a scale-invariant measure that isolates shape from size. For consecutive integers, $G/H \sim \log(n)/e$ exactly via Stirling's approximation. For consecutive primes, first-order analytic predictions fail dramatically: the naïve asymptotic $G/H \sim \log \log n$ underestimates the true value by a factor of three at $n = 1000$. Empirically, $G/H \approx C(n) \cdot \log p_n$ with $C(n) \approx 0.7$ increasing slowly, but no closed form for $C(n)$ is known. This contrast between exact and empirical regimes reflects a deeper structural divide: factorials admit Stirling's clean approximation while primorials do not, and the Euler–Mascheroni constant γ that governs integer asymptotics pervades physics, while the Meissel–Mertens constant B_1 that governs prime asymptotics has no known physical applications.

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1 What shape is a box?

The question in plain language

Take the integers 1, 2, 3 as the sides of a box. You get a $1 \times 2 \times 3$ cuboid with volume 6. Now take 1, 2, 3, 4 and build in four dimensions; the “box” has volume 24. In n dimensions with sides 1 through n , the volume is $n!$ and the shape grows steadily more elongated. Can we measure *how* elongated, independent of scale? And what changes if we replace consecutive integers with consecutive primes?

The concept of an *oblong number* as the product $n(n+1)$ of two consecutive integers dates to classical Greek arithmetic [1]. Generalising to n dimensions, the hyperrectangular solid with sides $1, 2, \dots, n$ has volume $n!$, linking combinatorics directly to geometry. We call this the **oblong cuboid**.

The natural question is whether we can characterise its *shape* as the proportional relationship among its sides, independently of its *size*. The surface-to-volume ratio S/V is the obvious candidate, but it fails: doubling all sides halves S/V . We need a scale-invariant quantity.

1.1 The isoperimetric quotient

For an n -dimensional cuboid with sides $a_1, \dots, a_n$, the surface area satisfies

$$S = 2V \sum_{i=1}^n \frac{1}{a_i} = \frac{2nV}{H}$$

where $H = n / \sum (1/a_i)$ is the harmonic mean. This is scale-dependent. The isoperimetric quotient removes the scale:

Theorem 1.1 (Isoperimetric Quotient for Cuboids). *For an n -dimensional cuboid with sides $a_1, \dots, a_n$:*

$$M = \frac{S}{V^{(n-1)/n}} = 2n \cdot \frac{G}{H}$$

where $G = (a_1 \cdots a_n)^{1/n}$ is the geometric mean and $H = n / \sum (1/a_i)$ is the harmonic mean.

Proof. From $S = 2nV/H$ and $V^{(n-1)/n} = V/V^{1/n} = V/G$:

$$M = \frac{S}{V^{(n-1)/n}} = \frac{2nV/H}{V/G} = 2n \cdot \frac{G}{H}. \quad \square$$

The factor $2n$ is the value of M for a hypercube (where $G = H$). Everything interesting about shape lives in the ratio G/H .

The shape measure

The ratio G/H is the fundamental, scale-invariant measure of side-length inequality:

- $G/H = 1$ if and only if all sides are equal (hypercube);
- $G/H > 1$ for any anisotropic cuboid;
- larger G/H indicates greater disparity among side lengths.

This follows from the AM–GM–HM inequality [2]: $H \leq G \leq A$, with equality iff all values coincide.

Our programme is now concrete: compute G/H as a function of dimension n , first for consecutive integers (where exact results are available), then for consecutive primes (where they are not).

2 Oblong cuboids: the exact answer

Definition 2.1 (Oblong Cuboid). The n -dimensional oblong cuboid has sides $1, 2, 3, \dots, n$, with volume $V = n!$, surface area $S = 2n! \cdot H_n^*$, geometric mean $G_n = (n!)^{1/n}$, and harmonic mean $H_n = n/H_n^*$, where $H_n^* = \sum_{i=1}^n 1/i$ is the n -th harmonic number.

Both means have clean asymptotics rooted in classical analysis.

Geometric mean. By Stirling’s approximation [2], $n! \sim \sqrt{2\pi n} (n/e)^n$, so

$$G_n = (n!)^{1/n} \sim \frac{n}{e}.$$

Harmonic mean. The Euler–Maclaurin formula [3] gives

$$H_n^* = \log n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + O(n^{-4})$$

where $\gamma = \lim_{n \rightarrow \infty} (H_n^* - \log n) \approx 0.5772$ is the Euler–Mascheroni constant [5]. Thus $H_n = n/H_n^* \sim n/\log n$.

Theorem 2.2 (Oblong Isoperimetric Quotient).

$$\frac{G_n}{H_n} \sim \frac{\log n}{e}$$

and the isoperimetric quotient $M_{\text{obl}}(n) = 2n \cdot G_n/H_n \sim 2n \log n/e$.

Why this is satisfying

The result is clean, closed-form, and exact to leading order. The Euler–Mascheroni constant, which enters via $\gamma = -\Gamma'(1)$, connects this geometry to the gamma function and thereby to regularisation procedures throughout physics, such as dimensional regularisation in QED [7], Feynman integrals, and analytic continuations of products and sums. The shape of a box made of integers speaks the same mathematical language as quantum field theory.

Shape asymmetry grows logarithmically: slowly, predictably, governed by constants that appear throughout mathematics and physics. We now ask what happens when regularity is replaced by the fundamental irregularity of the primes.

3 Puboids: where the theory breaks

Definition 3.1 (Puboid). The n -dimensional puboid has sides equal to the first n primes $p_1, p_2, \dots, p_n$. Its volume is the primorial $p_n\# = \prod_{i=1}^n p_i$.

We bring the same programme: compute G and H asymptotically, then take their ratio. Both computations require deep results from analytic number theory.

3.1 Geometric mean via Chebyshev

The geometric mean satisfies

$$G_n^{\text{pub}} = \exp\left(\frac{1}{n} \sum_{i=1}^n \log p_i\right) = \exp\left(\frac{\theta(p_n)}{n}\right)$$

where $\theta(x) = \sum_{p \leq x} \log p$ is Chebyshev's function. The Prime Number Theorem [3] gives $\theta(x) \sim x$, and since $p_n \sim n \log n$ (also PNT), we have

$$\frac{\theta(p_n)}{n} \sim \frac{p_n}{n} \sim \log n \implies G \sim \exp(\log n) = n.$$

3.2 Harmonic mean via Meissel–Mertens

The sum of reciprocal primes is controlled by the Meissel–Mertens theorem [3, 4]:

$$\sum_{i=1}^n \frac{1}{p_i} = \log \log p_n + B_1 + O\left(\frac{1}{\log p_n}\right)$$

where $B_1 = \gamma + \sum_p [\log(1 - 1/p) + 1/p] \approx 0.2615$ is the Meissel–Mertens constant. Thus

$$H \sim \frac{n}{\log \log p_n} \sim \frac{n}{\log \log n}.$$

3.3 The prediction and its failure

Combining first-order terms:

$$\frac{G}{H} \sim \frac{n}{n/\log \log n} = \log \log n.$$

The factor-of-three failure

At $n = 1000$, this predicts $G/H \approx 1.93$. The computed value is **6.07**—off by a factor of more than three.

This is not a small correction. The failure is structural, and understanding *why* it fails illuminates why prime asymptotics resist closed-form treatment.

3.4 Anatomy of the failure

The approximation $\theta(p_n) \sim p_n$ hides a slowly vanishing error: write $\theta(p_n) = p_n(1 - \varepsilon_n)$ where $\varepsilon_n \rightarrow 0^+$ at rate $\sim 1/\log p_n$. Then

$$G = \exp\left(\frac{p_n(1 - \varepsilon_n)}{n}\right) = \exp((1 - \varepsilon_n)(1 + \delta_n) \log n) = n^{(1 - \varepsilon_n)(1 + \delta_n)}$$

where $\delta_n = \log \log n / \log n + o(1/\log n)$ captures sub-leading terms in $p_n = n \log n(1 + \delta_n)$.

The exponential amplification

The error ε_n is “negligible” in the sense that $\theta(p_n)/p_n \rightarrow 1$. But it enters as an *exponent*: the geometric mean is $\exp(\theta(p_n)/n)$, so a small additive error in the exponent becomes a multiplicative error in G itself. Terms of order $\delta_n - \varepsilon_n$, though individually vanishing, compound through exponentiation to produce the observed factor-of-three discrepancy. This is the fundamental obstacle: **there is no “Stirling’s approximation for primorials”**.

A rigorous second-order analysis using the logarithmic integral $\text{Li}(x)$ or explicit error terms for $\theta(x)$ would improve predictions but is unlikely to yield closed forms, precisely because prime irregularity prevents the kind of clean asymptotic expansion that Stirling’s formula provides for factorials.

4 Empirical analysis

Since exact asymptotics fail, we turn to numerical computation. A sieve of Eratosthenes generates primes, and direct calculation of G and H via logarithmic sums gives:

n	p_n	Oblong G/H	$\log n/e$	Puboid G/H	$\log \log n$	$C(n)$
10	29	0.951	0.847	1.468	0.834	0.436
50	229	1.728	1.439	2.633	1.364	0.484
100	541	2.110	1.694	3.313	1.527	0.526
500	3571	3.131	2.286	5.181	1.827	0.634
1000	7919	3.545	2.540	6.072	1.933	0.676
5000	48611	4.618	3.136	8.271	2.149	0.767

Table 1: Comparison of oblong and puboid G/H ratios, with asymptotic predictions and the empirical coefficient $C(n) = (G/H)_{\text{pub}} / \log p_n$.

The data reveals a clear empirical law:

Conjecture 4.1 (Puboid G/H growth). For the puboid with sides $p_1, \dots, p_n$:

$$\frac{G}{H} \approx C(n) \cdot \log p_n$$

where $C(n)$ is a slowly increasing function satisfying $C(50) \approx 0.48$, $C(1000) \approx 0.68$, $C(5000) \approx 0.77$.

Remark 4.2 (Comparison of coefficients). The oblong formula $G/H \sim \log(n)/e$ has coefficient $1/e \approx 0.368$, while the puboid’s empirical coefficient $C(n) \approx 0.7$ is roughly 1.9 times larger. Primes stretch boxes almost twice as aggressively as integers do. For large n , $(G/H)_{\text{pub}} \approx 1.9 \cdot (G/H)_{\text{obl}}$.

5 Two constants, two worlds

The oblong and puboid asymptotics are governed by parallel constants that play analogous roles in correcting leading terms:

	Oblong (Integers)	Puboid (Primes)
Formula	$G/H \sim \log(n)/e$	$G/H \approx C(n) \cdot \log p_n$
Basis	Stirling's approximation	No analogue exists
Correcting constant	$\gamma \approx 0.577$	$B_1 \approx 0.261$
Asymptotic series	$\sum 1/i = \log n + \gamma + O(1/n)$	$\sum 1/p_i = \log \log p_n + B_1 + O(1/\log p_n)$
Closed coefficient	$1/e \approx 0.368$	$C(n) \approx 0.7$ (no closed form)

The constants are connected:

$$B_1 = \gamma + \sum_{p \text{ prime}} \left[\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right] \quad (1)$$

Both correct leading asymptotic terms in their respective summations. Yet their *universality* differs profoundly.

6 Why γ is physics and B_1 is not

The Euler–Mascheroni constant appears throughout physics: in QED regularisation (because $\gamma = -\Gamma'(1)$ and the gamma function provides analytic continuation for divergent integrals [7]), in statistical mechanics, in number-theoretic aspects of string amplitudes. The Meissel–Mertens constant does not appear in any known physical context. This is not an accident; it reflects four structural incompatibilities between prime distributions and physical theories:

- 1. Continuous symmetry.** Physical laws arise from continuous symmetries—Lorentz, gauge, diffeomorphism. Prime distributions exhibit discrete irregularity with no natural Lie group structure.
- 2. Density.** QED regularisation sums over *all* momentum states. Primes are too sparse ($p_n \sim n \log n$ vs. integers growing as n) to serve as a regularisation scheme.
- 3. Multiplicative vs. additive structure.** B_1 arises from unique factorisation—fundamentally multiplicative. Conservation laws (energy, momentum, charge) are additive. The mathematical structures do not match.
- 4. Analytic continuation.** $\Gamma(s)$ extends factorials; $\zeta(s)$ extends sums over integers. Primorials $\prod p_i$ and $\sum 1/p_i$ lack clean analytic continuations that could appear naturally in regularisation.

Universality vs. specificity

γ encodes the universal discrete-to-continuous transition: it measures how sums approximate integrals. B_1 encodes the specific structure of the primes within the integers: a precision matched to multiplicative number theory but alien to current physical frameworks. The isoperimetric quotient makes this distinction geometrically visible: identical measurement (G/H), qualitatively different behaviour.

7 Open problems and future directions

The nature of $C(n)$

Can the empirical coefficient $C(n)$ in $G/H \approx C(n) \cdot \log p_n$ be expressed via known number-theoretic functions? Natural candidates include:

- The logarithmic integral $\text{Li}(x) = \int_2^x dt/\log t$;
- Riemann's prime-counting approximation $R(x)$;
- Higher-order terms in $\theta(x) - x$.

Specifically, is there an analytic relationship between $C(n)$ and the remainder $p_n/\pi(p_n) - \log p_n$?

Information-theoretic interpretation

The surface-to-volume ratio $S/V = 2n/H$ measures boundary complexity relative to bulk. For anisotropic cuboids, M grows logarithmically rather than linearly, dramatically reducing this ratio compared to hypercubes. The cuboid's even slower growth ($\sim \log \log n$ at leading order) reflects prime sparsity's geometric manifestation. Can this be formalised as an entropy measure on integer sequences?

Beyond primes

What G/H growth rates arise from other arithmetic sequences such as twin primes, primes in arithmetic progressions, or the values of multiplicative functions? The isoperimetric quotient framework applies to any positive sequence; different growth rates would reflect different distributional properties.

8 Conclusions

The isoperimetric quotient $M = 2n \cdot G/H$ provides a scale-invariant shape measurement for n -dimensional cuboids. The ratio G/H directly quantifies side-length inequality, and its growth with dimension reveals fundamental properties of the underlying sequence:

$$\text{Oblong: } G/H \sim \frac{\log n}{e} \quad (\text{exact, via Stirling}) \quad (2)$$

$$\text{Puboid: } G/H \approx C(n) \cdot \log p_n \quad (\text{empirical, } C \approx 0.7) \quad (3)$$

This difference is not merely quantitative. Rather it reflects the structural divide between integer regularity which admits Stirling's clean asymptotic expansion, and prime

irregularity, which resists closed-form treatment at every turn. The Euler–Mascheroni constant γ and the Meissel–Mertens constant B_1 play parallel corrective roles in their respective asymptotic series, but differ fundamentally in universality: γ pervades physics through $\Gamma'(1)$, while B_1 remains confined to multiplicative number theory.

The isoperimetric quotient framework makes this abstract distinction geometrically concrete. The same measurement, applied to two natural sequences, produces one exact formula and one empirical law. The gap between them is a geometric shadow of the Prime Number Theorem’s irreducible error.

An interactive HTML companion provides a scroll-based walkthrough of the oblong–puboid construction, the isoperimetric quotient $M = 2n(G/H)$, and the empirical divergence between integer- and prime-sided cuboids [8].

References

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